

Report

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June 1, 2026

Abstract

Using CompAct, a software capable of solving systems of differential equations between compartments, the activity released into the environment following an incident sequence was investigated.

1 Introduction

To validate the modeling approach, a known accident with existing results was re-created using CompAct. This scenario is a LOCA in the chemical and volume control system (CVCS) outside of the secondary containment. The leak released contaminated water in the auxiliary building, the liquid part will generate a sump while the steam and volatile part will go in the atmosphere of the auxiliary building and ultimately in the environment. As the LOCA occurs, the unit gets shut down (SCRAM or in german RESA) and later, the emergency core cooling system (ECCS) starts refilling the primary circuit. In the case of the CVCS line, the isolation of the leakage occurs after 30 minutes (time to have operator action). In the analysis it is also distinguished between single failure assumed and not assumed.

2 Features

2.1 Initial Activities

The activity concentrations of the investigated reactor are measured so then are known. However they need to be scaled by a factor such that the concentration of I^{131} will result to be $2 \cdot 10^6 Bq/kg$, as instructed by the normative. Furthermore the different iodine have to be multiplied by their relative amounts: aerosol (95%), element (4.85%) and organic (0.15%). Since the input has to be activities and not concentrations they must multiplied by the mass of the primary circuit (see fig. 12,14 and 15).

2.2 Mass Conservation

The starting nuclei's activities must be multiplied for a considerable large factor ($UF = 10^6$) to guarantee the mass conservation from the source, which is the primary circuit. This way it's possible to have constant outflowing activity. Multiplying by a large factor such as 10^6 (named UF) to increase the initial activity out of scale. This will make the decrease of water negligible (see fig. 13). The possible decrease of concentration of nuclides in the leaking water is also ignored (conservative assumption).

2.3 Leaking Flow

2.3.1 UF

Since the leaking flow from the primary circuit follows the following equation: $\frac{dx}{dt} = Jx + \dots$ where "x" is the activity from the primary circuit and "J" the flow rate calculated as the ratio between mass flow and total mass of the primary circuit. Then the value of "x" will be UF times higher, so it's necessary to divide "J" by UF to have the correct value of the leaking flow (see fig. 17). As the primary circuit mass is constant and so is the activity. Thus, also the leaking flow doesn't decrease.

2.3.2 Spiking

As the station get shut down (RESA happens) the activity releases into the primary circuit increase. This phenomena is modeled by introducing a linear spike from the primary circuit to the auxiliary building. This spike will increase the releases of 25 times in 4.5 hours. Thus the flow rate will be as it was between "PKLpreRESA" and "Atm" but with an extra factor such as $(1 + 24(\frac{t-t_{RESA}}{4.5h}))$ (see fig. 17). Before the time of the RESA the leaking happens from a compartment representing the primary circuit before RESA (called "PKLpreRESA") where decay doesn't happen. At time of RESA the whole amount of activity is transported instantaneously (quick released rate= $1s^{-1} = 3600h^{-1}$) to another compartment representing the primary circuit post RESA (called "PKLpostRESA") where decay does happen (see fig. 18), and the leaking is increased following the spike. The time of RESA was taken as 0.01 hours following the model of Framatome; However, 0.01 hours corresponds to 36 seconds, which does not appear to be technically justified. To adopt a conservative approach, this time can be set to zero, resulting in an approximate 2% ¹ increase in the final released activity to the environment. For the validation model, the RESA time was maintained at 0.01 hours to ensure consistency with existing results, while in the new model, it will be set to zero.

2.4 Gasses and Liquid

It is known the mean enthalpy of the leaking water. In case of no single failure the enthalpy is low, so no steam is generated. Nevertheless, it is assumed that 10% of the iodine aerosol and 0.02% of the water will flash into steam. So, these together with the whole amount of noble gasses, element iodine and organic iodine go to the compartment representing the atmosphere of the auxiliary building (called "Atm"). While all the other nuclides go to the sump. When single failure is considered, it is added a certain amount of leaking water, with its steam fraction calculated from the respective enthalpy balance. The steam will go directly to the "Atm", the rest to the sump. (see fig. 17)

2.5 Sump

The sump is divided in three compartment: "fl-leakage", "sumpf" and "sumpf-I".

2.5.1 fl-leakage

"fl-leakage" is a support compartment where the liquid leakage goes and then get separated into the two different compartment: one ("sumpf-I") for the volatile aerosol iodine and noble gasses and the other ("sumpf") for everything else (aerosols) (see fig. 16).

2.5.2 sumpf

"sumpf" host 95% of the aerosol iodine, from here eventual generated noble gasses go to the compartment "sumpf-I" instantaneously (rate= $3600h^{-1}$).

2.5.3 sumpf-I

"sumpf-I" hosts 5% of the aerosol iodine which gets converted into elemental and organic iodine with different velocities to maintain the ratio $\frac{I_{el}}{I_{or}} = \frac{4.85}{0.15}$. So the velocities will be from aerosol to elemental iodine $3600 \cdot \frac{4.85}{5} h^{-1}$, and from aerosol to organic iodine $3600 \cdot \frac{0.15}{5} h^{-1}$. From "sumpf-I" to "Atm" it is set instantaneous release of noble gases, organic and elemental iodine (rate= $3600h^{-1}$). (see fig. 19 and 20).

2.6 Atm

From the "Atm" compartment, a deposition of the iodine element into the sump is defined in such a way as to maintain an activity concentration ratio of 100 (partition coefficient P). Thus, considering that the release velocity of element iodine from "sumpf-I" to "atm" was $3600h^{-1}$ the deposition will

¹ Activity = $\int_0^{t_{end}} Jx(1 + \frac{24}{4.5}(t - t_{RESA})) dt = Jx(t_{end} + 5.3(\frac{t_{end}^2}{2} - t_{end}t_{RESA}))$
 $A_0(t_{RESA} = 0h, t_{end} = 0.5h) = 1.1625, A_1(t_{RESA} = 0.01h, t_{end} = 0.5h) = 1.136 \rightarrow \frac{A_0}{A_1} = 102\%$

be $3600 \cdot 100 \cdot \frac{\text{volume sump} - I}{\text{volume Atm}} h^{-1}$. Furthermore, since in the auxiliary building there are no filters it is modeled that all the gasses are released in the environment with a rate equal to the volume flow divided by 50% of the volume of the auxiliary building. In fact the normatives specify that to use the equations implemented by CompAct, which are based on the assumption of homogeneous mixture of nuclides, when this assumption is not fulfilled it is necessary to use only half of the free volume. The volume of the sump is calculated as the mass of the water in the primary circuit ($194300kg$) divided by water density ($1000kg/m^3$). The volume of atm is then considered as the actual volume of the auxiliary building divided by two minus the volume of the sump. (see fig. 22 and 23)

3 Compare results

Framatome provided the results of this case, so it was possible to check the released activities for each nuclide during the time intervals 0h-8h, 8h-24h, 24h-72h and 72h-168h. The relative errors (between the calculated results and Framatome's), for the total emissions and for the first 8 hours, where the absolute emissions are higher, are almost always below 1%.

$$\text{Relative Error} = \frac{\text{calculated_activity} - \text{given_activity}}{\text{given_activity}}$$

Note: in the following sections if a graph doesn't show all the nuclides it means that it is showing only those with relative error $\neq 0$; I-131, Xe-133 and Cs-137 are highlighted.

3.1 Without single failure

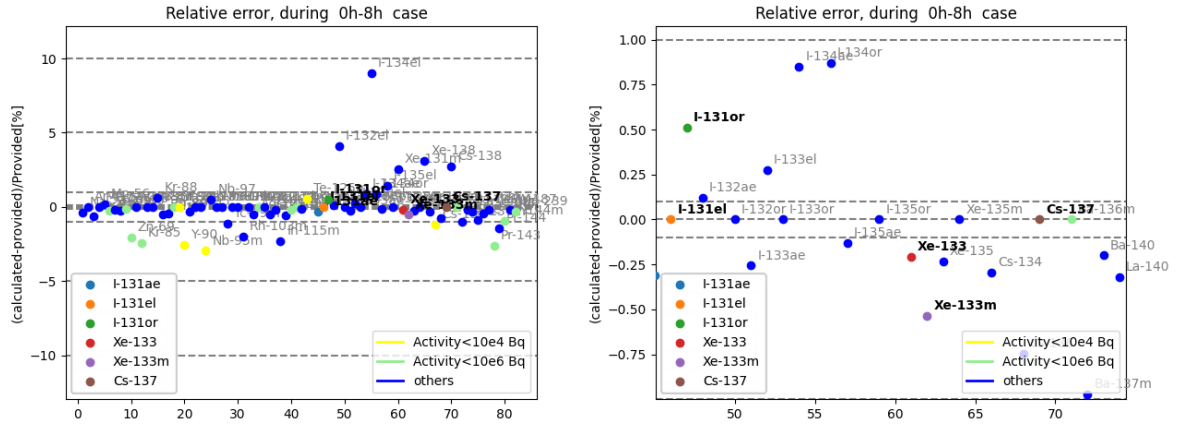


Figure 1: Releases in first 8 hour and zoomed version.

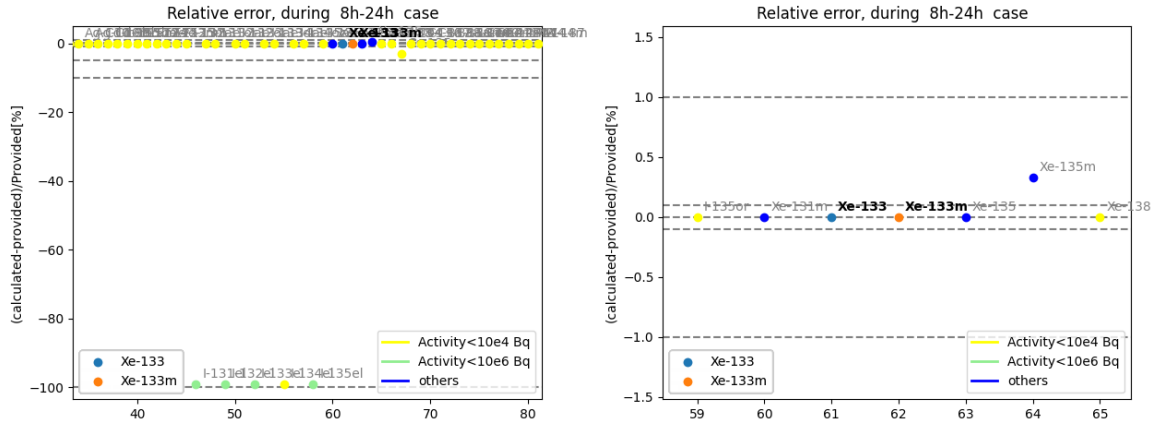


Figure 2: Releases between first 8-th hour and 24-th hour.

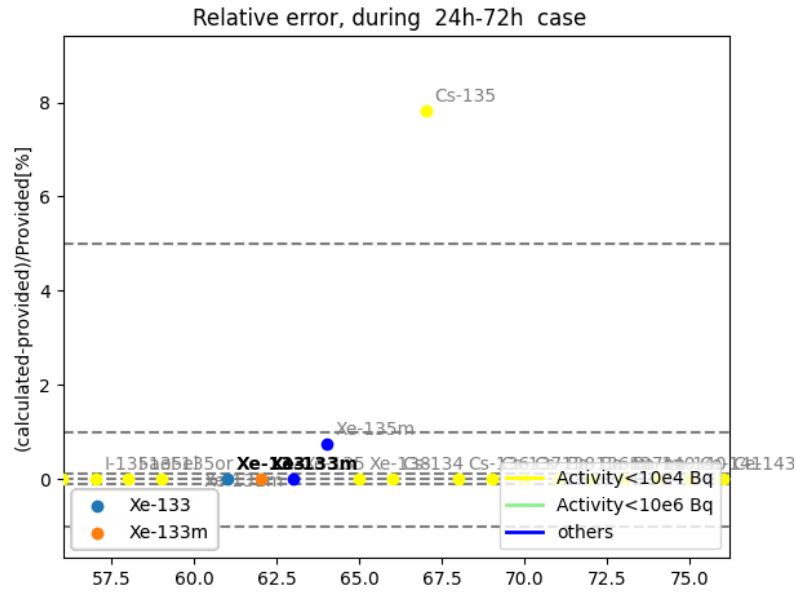


Figure 3: Releases between first 24-th hour and 72-th hour.

3.2 With single failure

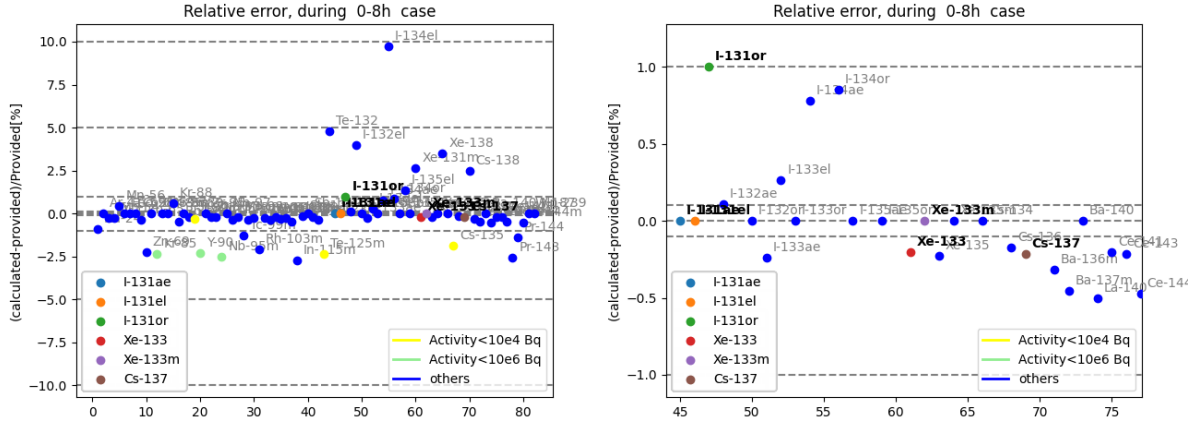


Figure 6: Releases in first 8 hour and zoomed version.

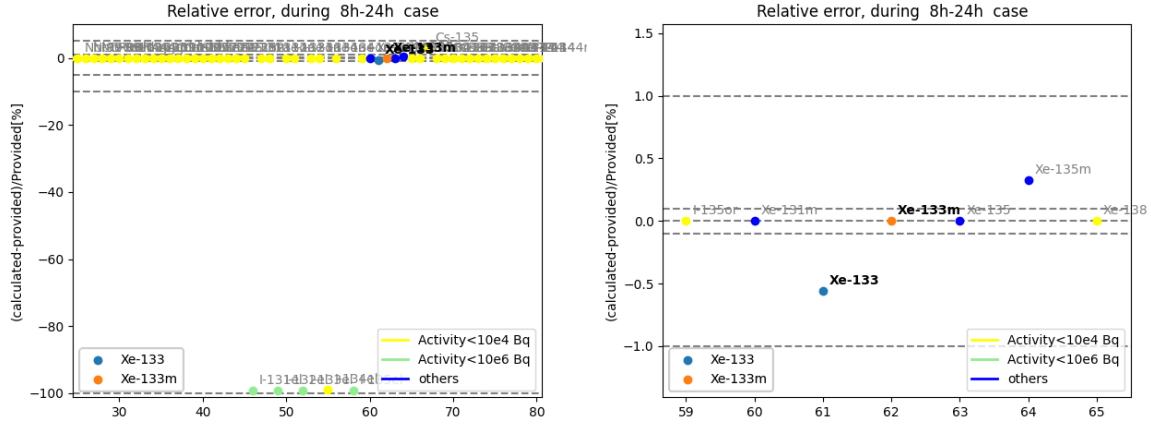


Figure 7: Releases between first 8-th hour and 24-th hour and zoomed version.

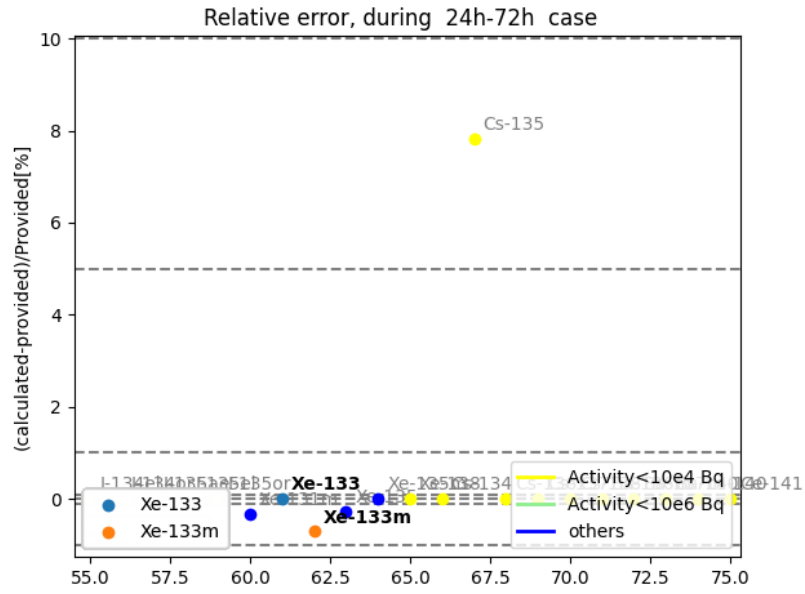


Figure 8: Releases between first 24-th hour and 72-th hour.

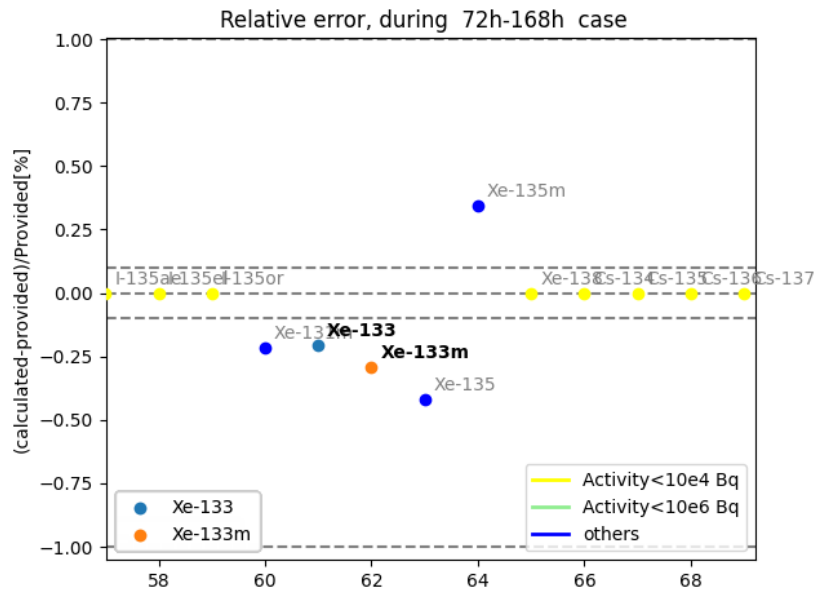


Figure 9: Releases between first 72-th hour and 168-th hour.

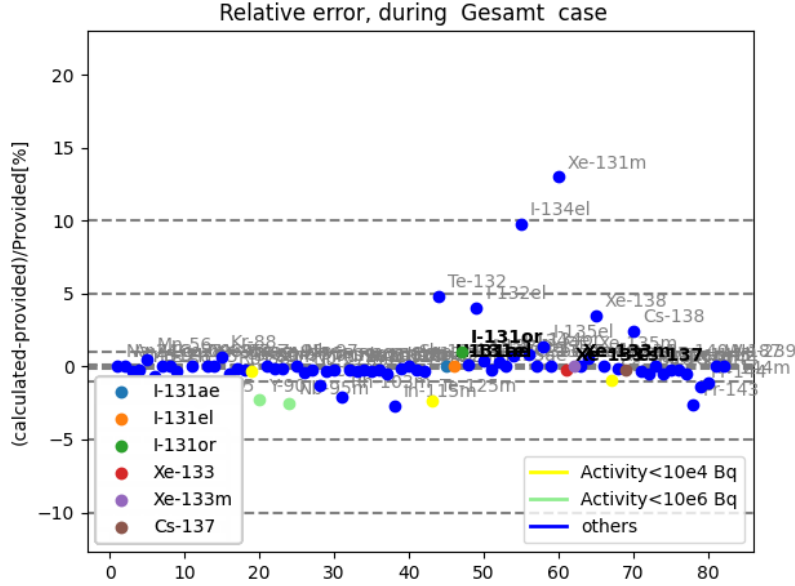


Figure 10: Total releases.

4 Dose

Even if the majority of the nuclides have relative error lower than 1% what really matters is the dose exposure of the public. So it was necessary to prepare and validate the setting of EXPOG14, software able to calculate the dose related to a certain activity released. Thus using the activity released Framatome provided and assuring that the resulting doses were the same they provided, it was known that the setting of EXPO14 was correct. Finally the dose was calculated starting from the activities output of the implemented model and compared them with Framatome's ones.

To calculate the dose, weather and topological data have to be loaded in EXPO14, in this both cases: LOCA in the auxiliary building and in the reactor building, the gasses go in the environment through the stack. The release is assumed at 99m, the building effect is considered only for the cooling tower (which is 150m high).

Doses without Single Failure

Table 1: Radiation doses, provided by Framatome, in Sector 20 for different age groups

Dose [mSv]	Infant	10-year-old Child	Adult
Max. Cloud Dose (Sector 20)	1.57e-01	1.15e-01	9.79e-02
Max. Ground Dose (Sector 20)	3.90e+00	1.47e+00	7.05e-01
Effective Total Dose (Sector 20)	4.06e+00	1.58e+00	8.02e-01

Table 2: Radiation doses in Sector 20 after a LOCA in the Auxiliary building (ohne EF)

Dose [mSv]	Infant	10-year-old Child	Adult
Maximum Cloud Dose	1.59e-01	1.16e-01	9.89e-02
Maximum Ground Dose	3.86e+00	1.45e+00	6.99e-01
Effective Total Dose	4.02e+00	1.57e+00	7.98e-01

Table 3: relative error: $\frac{\text{calculated}-\text{provided}}{\text{provided}}$ on doses in Sector 20 after a LOCA in the Auxiliary Building (ohne EF)

dose increase	Infant	10-year-old Child	Adult
Maximum Cloud Dose	0.64%	0.87%	1.02%
Maximum Ground Dose	-1.03%	-1.36%	-0.85%
Effective Total Dose	-0.99%	-0.63%	-0.5%

Doses with Single Failure

Table 4: Radiation doses, provided by Framatome, in Sector 20 for different age groups

Dose [mSv]	Infant	10-year-old Child	Adult
Maximum Cloud Dose	1.70e-01	1.25e-01	1.08e-01
Maximum Ground Dose	4.17e+00	1.62e+00	8.07e-01
Effective Total Dose	4.34e+00	1.74e+00	9.15e-01

Table 5: Radiation doses in Sector 20 after a LOCA in the Auxiliary building (mit EF)

Dose [mSv]	Infant	10-year-old Child	Adult
Maximum Cloud Dose	1.71e-01	1.25e-01	1.09e-01
Maximum Ground Dose	4.17e+00	1.62e+00	8.09e-01
Effective Total Dose	4.34e+00	1.74e+00	9.18e-01

Table 6: relative error: $\frac{\text{calculated}-\text{provided}}{\text{provided}}$ on doses in Sector 20 after a LOCA in the Auxiliary Building (adult EF)

dose increase	Infant	10-year-old Child	Adult
Maximum Cloud Dose	+0.59%	+0.0%	+0.93%
Maximum Ground Dose	+0.0%	+0.0%	+0.25%
Effective Total Dose	+0.0%	+0.0%	+0.33%

As one can see, resulting doses are almost identical, thus the differences in activity releases are negligible. Therefore, the implemented model can be considered validated.

5 LOCA in Reactor Building

After validating the model we created another with a leak in the CVC letdown line inside the reactor building ("Ring Raum"). The main differences between the two LOCA are the values of some variables (see fig. 25) and that in the reactor building case after 30 min the filtered ventilation system is activated so the gasses' flux to the environment decrease and pass through filters. For this reason this reason it was simply modified the existing model.

5.1 Filtering

When the LOCA happens in the "Ring Raum", after 30 minutes, the leak is isolated and the filtered ventilation activated. By activation of the filtered ventilation the volume flow decrease from $60000m^3/h$ to $4000m^3/h$; no noble gas get trapped in the filter while 99.9% of element iodine, 99% of organic iodine and 99.9% of everything else get absorbed by the filter. The model assume that any noble gas generated by the radioactive decay of a nuclide absorbed in the filter is released into the environment instantaneously (flux rate $3600h^{-1}$) (see fig. 26)

6 Results RiRa

6.1 Activities RiRa without single failure

Nuclide Activity [Bq]	0-8h	8-24h	24-72h	72-168h
Na-24	3.34e+07	2.47e+03	7.80e+01	0.00e+00
Ar-41	2.38e+11	1.26e+09	1.85e+05	0.00e+00
Cr-51	2.17e+05	2.80e+01	2.00e+00	0.00e+00
Mn-54	2.47e+05	3.20e+01	2.00e+00	0.00e+00
Mn-56	6.96e+05	5.00e+00	0.00e+00	0.00e+00
Fe-59	4.55e+04	6.00e+00	0.00e+00	0.00e+00
Co-58	7.09e+05	9.10e+01	6.00e+00	0.00e+00
Co-60	5.66e+05	7.30e+01	5.00e+00	0.00e+00
Zn-65	8.93e+04	1.20e+01	1.00e+00	0.00e+00
Zn-69	7.17e+04	2.50e+01	1.00e+00	0.00e+00
Zn-69m	3.26e+05	2.30e+01	1.00e+00	0.00e+00
Kr-85	4.90e+05	3.41e+05	2.73e+04	7.00e+00
Kr-85m	1.57e+11	6.49e+09	3.50e+07	5.00e+00
Kr-87	2.52e+11	3.12e+08	3.25e+03	0.00e+00
Kr-88	3.50e+11	6.27e+09	8.07e+06	0.00e+00
Rb-88	4.92e+10	2.10e+06	2.70e+03	0.00e+00
Sr-89	3.04e+05	3.90e+01	3.00e+00	0.00e+00
Sr-90	1.47e+04	2.00e+00	0.00e+00	0.00e+00
Y-89m	2.90e+01	0.00e+00	0.00e+00	0.00e+00
Y-90	5.40e+01	0.00e+00	0.00e+00	0.00e+00
Zr-95	4.37e+06	5.62e+02	3.80e+01	0.00e+00
Zr-97	1.67e+06	1.31e+02	4.00e+00	0.00e+00
Nb-95	1.70e+07	2.18e+03	1.47e+02	0.00e+00
Nb-95m	1.29e+02	1.00e+00	0.00e+00	0.00e+00
Nb-97	5.71e+06	1.43e+02	5.00e+00	0.00e+00
Nb-97m	1.58e+06	1.25e+02	4.00e+00	0.00e+00
Mo-99	2.94e+07	3.34e+03	1.90e+02	0.00e+00
Tc-99m	4.08e+06	2.40e+03	1.75e+02	0.00e+00
Ru-103	1.79e+06	2.29e+02	1.50e+01	0.00e+00
Ru-106	2.41e+07	3.11e+03	2.12e+02	0.00e+00
Rh-103m	3.88e+05	2.26e+02	1.50e+01	0.00e+00
Rh-106	2.41e+07	3.11e+03	2.12e+02	0.00e+00
Ag-109m	2.58e+07	3.34e+03	2.27e+02	0.00e+00

Nuclide Activity [Bq]	0-8h	8-24h	24-72h	72-168h
Ag-110	2.67e+04	3.00e+00	0.00e+00	0.00e+00
Ag-110m	1.96e+06	2.54e+02	1.70e+01	0.00e+00
Cd-109	2.59e+07	3.34e+03	2.27e+02	0.00e+00
Cd-115	6.58e+06	7.25e+02	4.00e+01	0.00e+00
In-115m	3.37e+05	6.40e+02	4.20e+01	0.00e+00
Sb-122	2.33e+05	2.60e+01	2.00e+00	0.00e+00
Sb-124	6.36e+04	8.00e+00	1.00e+00	0.00e+00
Sb-125	1.34e+04	2.00e+00	0.00e+00	0.00e+00
Te-123m	8.84e+05	1.14e+02	8.00e+00	0.00e+00
Te-132	4.00e+06	4.62e+02	2.70e+01	0.00e+00
I-131ae	4.18e+09	5.17e+05	3.31e+04	7.00e+00
I-131el	3.64e+09	1.90e+06	1.62e+05	7.90e+01
I-131or	1.22e+08	5.08e+05	3.25e+04	7.00e+00
I-132ae	1.17e+11	5.78e+05	3.21e+02	0.00e+00
I-132el	1.01e+11	2.02e+06	1.34e+03	0.00e+00
I-132or	3.39e+09	5.67e+05	2.88e+02	0.00e+00
I-133ae	5.12e+10	4.41e+06	1.72e+05	9.00e+00
I-133el	4.46e+10	1.61e+07	8.26e+05	9.70e+01
I-133or	1.50e+09	4.34e+06	1.69e+05	9.00e+00
I-134ae	1.82e+11	1.03e+04	0.00e+00	0.00e+00
I-134el	1.58e+11	3.56e+04	0.00e+00	0.00e+00
I-134or	5.26e+09	1.01e+04	0.00e+00	0.00e+00
I-135ae	1.03e+11	3.89e+06	4.64e+04	0.00e+00
I-135el	8.98e+10	1.40e+07	2.18e+05	1.00e+00
I-135or	3.01e+09	3.82e+06	4.56e+04	0.00e+00
Xe-131m	1.14e+07	4.13e+07	1.18e+08	1.83e+08
Xe-133	1.35e+12	3.17e+11	1.06e+11	2.22e+10
Xe-133m	5.20e+10	1.34e+10	6.56e+09	1.48e+09
Xe-135	2.11e+12	8.51e+11	1.85e+11	1.17e+09
Xe-135m	8.82e+11	6.93e+11	1.71e+11	1.09e+09
Xe-138	4.89e+11	2.00e+00	0.00e+00	0.00e+00
Cs-134	4.44e+05	5.70e+01	4.00e+00	0.00e+00
Cs-135	3.00e+00	0.00e+00	0.00e+00	0.00e+00
Cs-136	1.74e+05	2.20e+01	1.00e+00	0.00e+00
Cs-137	1.43e+06	1.85e+02	1.30e+01	0.00e+00
Cs-138	7.41e+10	3.94e+02	0.00e+00	0.00e+00
Ba-136m	1.93e+04	2.00e+00	0.00e+00	0.00e+00
Ba-137m	1.31e+06	1.75e+02	1.20e+01	0.00e+00
Ba-140	6.51e+06	8.19e+02	5.40e+01	0.00e+00
La-140	1.19e+06	2.82e+02	2.80e+01	0.00e+00
Ce-141	1.52e+06	1.94e+02	1.30e+01	0.00e+00
Ce-143	2.84e+06	2.84e+02	1.40e+01	0.00e+00
Ce-144	6.43e+06	8.30e+02	5.60e+01	0.00e+00
Pr-143	2.06e+03	9.00e+00	1.00e+00	0.00e+00
Pr-144	3.46e+06	8.30e+02	5.60e+01	0.00e+00
Pr-144m	5.03e+04	8.00e+00	1.00e+00	0.00e+00
W-187	6.37e+06	5.79e+02	2.40e+01	0.00e+00
Np-239	4.45e+04	5.00e+00	0.00e+00	0.00e+00

Table 7: Nuclides activities with single failure

6.2 Activities RiRa with single failure

Nuclide Activity [Bq]	0-8h	8-24h	24-72h	72-168h
Na-24	3.56e+09	2.63e+05	8.27e+03	0.00e+00
Ar-41	2.56e+11	1.35e+09	1.99e+05	0.00e+00
Cr-51	2.31e+07	2.95e+03	1.97e+02	0.00e+00
Mn-54	2.63e+07	3.40e+03	2.31e+02	0.00e+00
Mn-56	7.41e+07	5.09e+02	0.00e+00	0.00e+00
Fe-59	4.85e+06	6.22e+02	4.20e+01	0.00e+00
Co-58	7.55e+07	9.71e+03	6.56e+02	0.00e+00
Co-60	6.03e+07	7.79e+03	5.30e+02	0.00e+00
Zn-65	9.51e+06	1.23e+03	8.30e+01	0.00e+00
Zn-69	7.64e+06	2.62e+03	7.70e+01	0.00e+00
Zn-69m	3.47e+07	2.45e+03	7.20e+01	0.00e+00
Kr-85	5.27e+05	3.67e+05	2.94e+04	8.00e+00
Kr-85m	1.69e+11	6.99e+09	3.77e+07	6.00e+00
Kr-87	2.71e+11	3.36e+08	3.50e+03	0.00e+00
Kr-88	3.77e+11	6.76e+09	8.68e+06	0.00e+00
Rb-88	5.44e+10	2.26e+06	2.91e+03	0.00e+00
Sr-89	3.24e+07	4.16e+03	2.81e+02	0.00e+00
Sr-90	1.57e+06	2.03e+02	1.40e+01	0.00e+00
Y-89m	3.12e+03	0.00e+00	0.00e+00	0.00e+00
Y-90	5.78e+03	2.60e+01	4.00e+00	0.00e+00
Zr-95	4.66e+08	5.99e+04	4.05e+03	1.00e+00
Zr-97	1.78e+08	1.40e+04	4.75e+02	0.00e+00
Nb-95	1.81e+09	2.32e+05	1.56e+04	4.00e+00
Nb-95m	1.37e+04	6.20e+01	9.00e+00	0.00e+00
Nb-97	6.08e+08	1.52e+04	5.12e+02	0.00e+00
Nb-97m	1.69e+08	1.33e+04	4.52e+02	0.00e+00
Mo-99	3.13e+09	3.55e+05	2.02e+04	3.00e+00
Tc-99m	4.35e+08	2.56e+05	1.87e+04	3.00e+00
Ru-103	1.90e+08	2.44e+04	1.64e+03	0.00e+00
Ru-106	2.57e+09	3.32e+05	2.25e+04	6.00e+00
Rh-103m	4.13e+07	2.41e+04	1.62e+03	0.00e+00
Rh-106	2.56e+09	3.32e+05	2.25e+04	6.00e+00
Ag-109m	2.75e+09	3.56e+05	2.42e+04	6.00e+00
Ag-110	2.84e+06	3.67e+02	2.50e+01	0.00e+00
Ag-110m	2.09e+08	2.70e+04	1.84e+03	0.00e+00
Cd-109	2.76e+09	3.56e+05	2.42e+04	6.00e+00
Cd-115	7.00e+08	7.72e+04	4.22e+03	1.00e+00
In-115m	3.58e+07	6.82e+04	4.52e+03	1.00e+00
Sb-122	2.48e+07	2.82e+03	1.60e+02	0.00e+00
Sb-124	6.77e+06	8.70e+02	5.90e+01	0.00e+00
Sb-125	1.43e+06	1.84e+02	1.30e+01	0.00e+00
Te-123m	9.41e+07	1.21e+04	8.23e+02	0.00e+00
Te-125m	5.60e+01	0.00e+00	0.00e+00	0.00e+00
Te-132	4.26e+08	4.92e+04	2.88e+03	0.00e+00
I-131ae	5.06e+09	6.25e+05	4.00e+04	9.00e+00
I-131el	3.89e+09	2.03e+06	1.73e+05	8.40e+01
I-131or	1.31e+08	5.43e+05	3.48e+04	8.00e+00
I-132ae	1.41e+11	7.48e+05	3.32e+03	0.00e+00
I-132el	1.08e+11	2.17e+06	1.44e+03	0.00e+00
I-132or	3.62e+09	6.07e+05	3.08e+02	0.00e+00
I-133ae	6.20e+10	5.34e+06	2.08e+05	1.10e+01
I-133el	4.77e+10	1.72e+07	8.84e+05	1.03e+02
I-133or	1.60e+09	4.64e+06	1.80e+05	9.00e+00

Nuclide Activity [Bq]	0-8h	8-24h	24-72h	72-168h
I-134ae	2.20e+11	1.25e+04	0.00e+00	0.00e+00
I-134el	1.69e+11	3.80e+04	0.00e+00	0.00e+00
I-134or	5.63e+09	1.09e+04	0.00e+00	0.00e+00
I-135ae	1.25e+11	4.71e+06	5.62e+04	0.00e+00
I-135el	9.61e+10	1.49e+07	2.34e+05	1.00e+00
I-135or	3.22e+09	4.09e+06	4.88e+04	0.00e+00
Xe-131m	1.22e+07	4.43e+07	1.26e+08	1.96e+08
Xe-133	1.46e+12	3.41e+11	1.14e+11	2.38e+10
Xe-133m	5.60e+10	1.44e+10	7.03e+09	1.59e+09
Xe-135	2.27e+12	9.13e+11	1.98e+11	1.25e+09
Xe-135m	9.66e+11	7.73e+11	1.91e+11	1.22e+09
Xe-138	5.26e+11	2.00e+00	0.00e+00	0.00e+00
Cs-134	4.73e+07	6.11e+03	4.15e+02	0.00e+00
Cs-135	4.00e+00	0.00e+00	0.00e+00	0.00e+00
Cs-136	1.85e+07	2.33e+03	1.53e+02	0.00e+00
Cs-137	1.52e+08	1.97e+04	1.34e+03	0.00e+00
Cs-138	1.17e+11	4.71e+02	0.00e+00	0.00e+00
Ba-136m	2.05e+06	2.58e+02	1.70e+01	0.00e+00
Ba-137m	1.40e+08	1.86e+04	1.27e+03	0.00e+00
Ba-140	6.94e+08	8.72e+04	5.71e+03	1.00e+00
La-140	1.27e+08	3.01e+04	3.00e+03	1.00e+00
Ce-141	1.62e+08	2.07e+04	1.39e+03	0.00e+00
Ce-143	3.02e+08	3.02e+04	1.44e+03	0.00e+00
Ce-144	6.85e+08	8.84e+04	6.01e+03	2.00e+00
Pr-143	2.20e+05	9.11e+02	1.21e+02	0.00e+00
Pr-144	3.68e+08	8.84e+04	6.01e+03	2.00e+00
Pr-144m	5.35e+06	8.44e+02	5.70e+01	0.00e+00
W-187	6.78e+08	6.17e+04	2.58e+03	0.00e+00
Np-239	4.73e+06	5.26e+02	2.90e+01	0.00e+00

Table 8: Nuclides activities with single failure

6.3 Dose RiRa

Without Single Failure

Table 9: Radiation doses in Sector 20 for different age groups

Dose [mSv]	Infant	10-year-old Child	Adult
Maximum Ground Dose	4.67e-01	1.76e-01	8.50e-02
Maximum Cloud Dose	2.06e-02	1.53e-02	1.32e-02
Effective Total Dose	4.87e-01	1.91e-01	9.81e-02

With Single Failure

Table 10: Radiation doses in Sector 20 for different age groups

Dose [mSv]	Infant	10-year-old Child	Adult
Maximum Ground Dose	5.52e-01	2.25e-01	1.18e-01
Maximum Cloud Dose	2.49e-02	1.88e-02	1.66e-02
Effective Total Dose	5.77e-01	2.43e-01	1.34e-02

A Appendix: Images

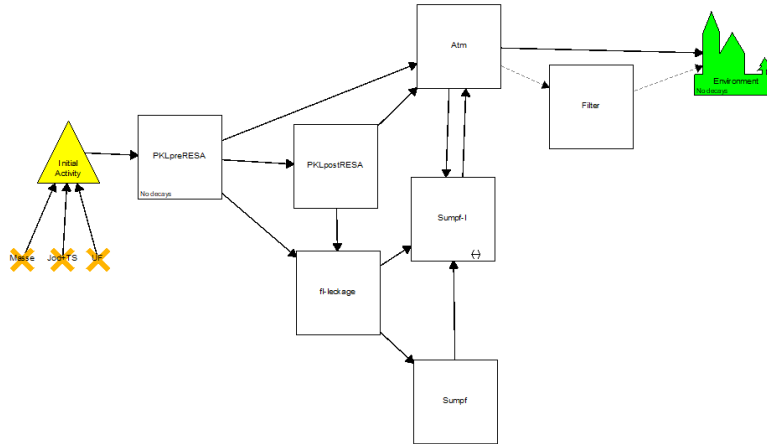


Figure 11: total scheme

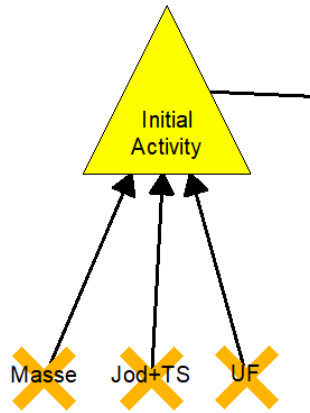


Figure 12: scheme for initial activity

UF

Item type: Multiplier

☒ Scale by Group
☐ Scale by Isotope

Group Scaling Factors Isotope Scaling Factors










ID	Group scaling Factor
1 Default	=UF

Figure 13: Multiply everything for UF

Jod+TS

Item type: Multiplier

☒ Scale by Group
☐ Scale by Isotope

Group Scaling Factors Isotope Scaling Factors










ID	Group scaling Factor
1 Default	=TS
2 iodine_aerosol	=lae*TS
3 iodine_element	=lel*TS
4 iodine_organic	=lor*TS

Figure 14: Adjust Iodine and scale everything $TS = \frac{2 \cdot 10^6 Bq/kg}{Activity_{I131}}$

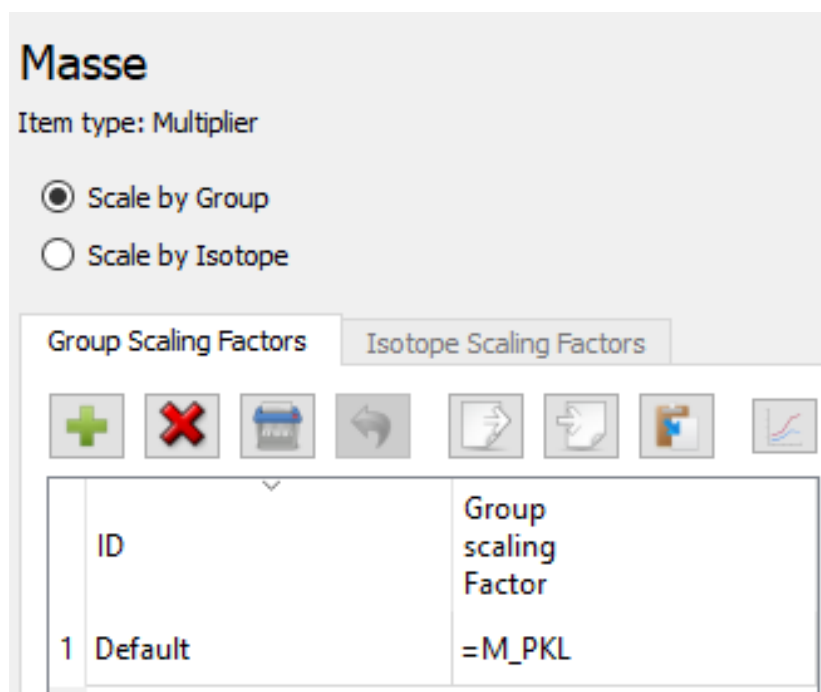


Figure 15: Multiply everything for the primary circuit mass to obtain the activity from the concentration.

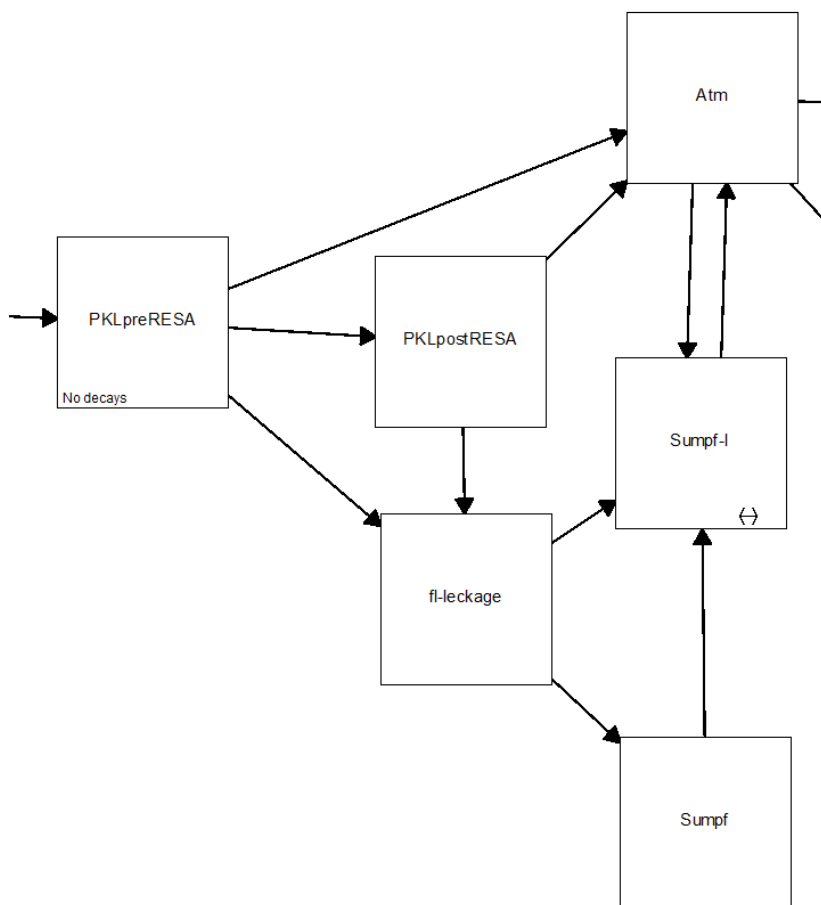


Figure 16: Scheme for atmosphere of auxiliary building and sump.

PKLpreRESA → Atm

Item type: Connection

☐ Advanced filter treatment

Flow Rates Filter Rates Filter Release Rates










ID	Start Time [h]	End Time [h]	Flow rate [1/h]
1 Default	t_{RESA}	t_{end}	0.0
2 noble_gas	t_{start}	t_{RESA}	$= R \cdot (Q_{ex} \cdot EF + Q_m) / M_{PKL} / UF; 1; lin$
3 iodine_element	t_{start}	t_{RESA}	$= R \cdot (Q_{ex} \cdot EF + Q_m) / M_{PKL} / UF; 1; lin$
4 iodine_organic	t_{start}	t_{RESA}	$= R \cdot (Q_{ex} \cdot EF + Q_m) / M_{PKL} / UF; 1; lin$
5 iodine_aerosol	t_{start}	t_{RESA}	$= R \cdot (F_{ex} \cdot Q_{ex} \cdot EF + F_{ae} \cdot Q_m) / M_P \dots$
6 Default	t_{start}	t_{RESA}	$= R \cdot (F_{ex} \cdot Q_{ex} \cdot EF + F \cdot Q_m) / M_{PKL} \dots$

Figure 17: The rest of the mass flow is transported between "PKLpreRESA" and "fl-leakage" (0 for NG, I_{el}, I_{or} and $(1 - F) \times (...)$ for Default and I_{ae}). From "PKLpostRESA" to "Atm" and "fl-leakage" the flows are the same but with the extra factor $(1 + 24 \frac{t - t_{RESA}}{4.5h})$.

PKLpreRESA → PKLpostRESA

Item type: Connection

☐ Advanced filter treatment

Flow Rates Filter Rates Filter Release Rates










ID	Start Time [h]	End Time [h]	Flow rate [1/h]
1 Default	t_{RESA}	t_{end}	3'600.00000000

Figure 18: Quick transfer of everything at $t = t_{RESA}$ ($rate = 3600h^{-1}$)

fl-leakage → Sumpf-I

Item type: Connection

☐ Advanced filter treatment

Flow Rates Filter Rates Filter Release Rates










ID	Start Time [h]	End Time [h]	Flow rate [1/h]
1 Default	t_{start}	t_{end}	0.0
2 iodine_aerosol	t_{start}	t_{end}	$= 3600 \cdot F_{vi}; 1; lin$
3 noble_gas	t_{start}	t_{end}	3'600.00000000

Figure 19: "Fvi" volume fraction volatile iodine ($F_{vi}=5\%$). (from "fl-leakage" to "Sumpf" is $3600 \cdot (1 - F_{vi})$ for I_{ae} , 3600 for default and 0 for NG).

Sumpf-I
Item type: Compartment/Environment
☒ Decays on

Depositions Conversions

	ID	ID2	Start Time [h]	End Time [h]	Conversion rates [1/h]
1	I-131 (ae)	I-131 (el)	t_start	t_end	$=3600 \cdot \text{el} / (\text{el} + \text{lor}); 1; \text{lin}$
2	I-131 (ae)	I-131 (or)	t_start	t_end	$=3600 \cdot \text{lor} / (\text{el} + \text{lor}); 1; \text{lin}$
3	I-132 (ae)	I-132 (el)	t_start	t_end	$=3600 \cdot \text{el} / (\text{el} + \text{lor}); 1; \text{lin}$
4	I-132 (ae)	I-132 (or)	t_start	t_end	$=3600 \cdot \text{lor} / (\text{el} + \text{lor}); 1; \text{lin}$
5	I-133 (ae)	I-133 (el)	t_start	t_end	$=3600 \cdot \text{el} / (\text{el} + \text{lor}); 1; \text{lin}$
6	I-133 (ae)	I-133 (or)	t_start	t_end	$=3600 \cdot \text{lor} / (\text{el} + \text{lor}); 1; \text{lin}$
7	I-134 (ae)	I-134 (el)	t_start	t_end	$=3600 \cdot \text{el} / (\text{el} + \text{lor}); 1; \text{lin}$
8	I-134 (ae)	I-134 (or)	t_start	t_end	$=3600 \cdot \text{lor} / (\text{el} + \text{lor}); 1; \text{lin}$
9	I-135 (ae)	I-135 (el)	t_start	t_end	$=3600 \cdot \text{el} / (\text{el} + \text{lor}); 1; \text{lin}$
10	I-135 (ae)	I-135 (or)	t_start	t_end	$=3600 \cdot \text{lor} / (\text{el} + \text{lor}); 1; \text{lin}$

Figure 20: Conversion of aerosol iodine into element and organic iodine.

Atm → Sumpf-I
Item type: Connection
☐ Advanced filter treatment

Flow Rates Filter Rates Filter Release Rates

ID	Start Time [h]	End Time [h]	Flow rate [1/h]
1 iodine_element	t_start	t_end	$=\text{vd} \cdot \text{P} \cdot \text{Vw} / (\text{V_cont} \cdot \text{VF} - \text{Vw}); 1; \text{lin}$

Figure 21: Deposition, the ratio of concentration of activities between "Sumpf-I" and "atm" has to be 100 (partition equilibrium).

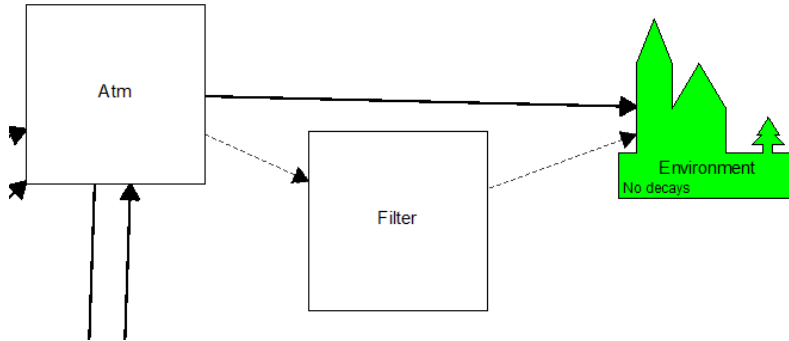


Figure 22: Scheme for release from the atmosphere of the auxiliary building. In this case the filter is not present.

Atm → Environment
Item type: Connection
☐ Advanced filter treatment

Flow Rates Filter Rates Filter Release Rates

ID	Start Time [h]	End Time [h]	Flow rate [1/h]
1 Default	t_start	t_end	$=\text{Qv} / (\text{V_cont} \cdot \text{VF} - \text{Vw}); 1; \text{lin}$

Figure 23: Release rate into the environment.

Global Parameters	
Mass conservation(UF)	1.000000000000E+06
Spike(SF)	25.00
Equilibrium Element(P)	100.00
Deposition Speed(vd)	3600.00
Aerosol Iodine fraction(Iae)	0.950000000000
Element Iodine fraction(Iel)	0.048500000000
Organic Iodine fraction(Ior)	0.001500000000
Volume Containment(V_cont)	39700.00
Volume Flow(Qv)	70000.00
Vfrac(VF)	0.500000000000
Mass PKL(M_PKL)	194300.00
Mass Flow(Qm)	148000.000000
Mass Flow Extra(Qex)	3770.00000000
Volume Water(Vw)	194.30
Steam Fraction(F)	2.000000000000E-04
Steam Fraction extra mass(Fex)	0.328381963000
Steam Fraction Iodine_ae F(Fae)	0.100000000000
volatile Iodine(Fvi)	0.050000000000
Time RESA(t_RESa)	0.01
EF logical(EF)	☒
Release before RESA(R)	☒
Scaling(TS)	20.2900000000

Figure 24: All variable used for LOCA in auxiliary building.

Global Parameters	
Mass conservation(UF)	1.000000000000E+06
Spike(SF)	25.00
Equilibrium Element(P)	100.00
Diffusion speed(vd)	3600.00
Aerosol Iodine fraction(Iae)	0.950000000000
Element Iodine fraction(Iel)	0.048500000000
Organic Iodine fraction(Ior)	0.001500000000
Volume Containment(V_cont)	46500.00
Volume Flow(Qv)	60000.00
Volume Flow Filtered(Qvf)	4000.00
Vfrac(VF)	0.500000000000
Mass PKL(M_PKL)	194300.00
Mass Flow(Qm)	50000.000000
Mass Flow Extra(Qex)	3828.00000000
Volume Water(Vw)	25.00
Steam Fraction(F)	2.000000000000E-04
Steam Fraction extra mass(Fex)	0.27553814002
Steam Fraction Iodine_ae F(Fae)	0.100000000000
Volatile Fraction I_ae(Fvi)	0.050000000000
Time RESA(t_RESa)	0.00
EF logical(EF)	☐
Release before RESA(R)	☒
Scaling(TS)	20.2839756592
Filtering(FF)	0.999000000000
Filtering orgaic(FFor)	0.990000000000
Filtering element(FFel)	0.999000000000

Figure 25: All variable used for LOCA in reactor building.

RiRa → Environment

Item type: Connection

☐ Advanced filter treatment

Flow Rates

Filter Rates

Filter Release Rates

+

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ID	Start Time [h]	End Time [h]	Flow rate [1/h]
1 Default	t_end_leak	t_end	$= (1 - FF) * Q_{vf} / (V_{cont} * VF); 1; lin$
2 iodine_organic	t_end_leak	t_end	$= (1 - FF_{or}) * Q_{vf} / (V_{cont} * VF); 1; lin$
3 iodine_element	t_end_leak	t_end	$= (1 - FF_{el}) * Q_{vf} / (V_{cont} * VF); 1; lin$
4 noble_gas	t_end_leak	t_end	$= Q_{vf} / (V_{cont} * VF); 1; lin$
5 Default	t_start	t_end_leak	$= Q_v / (V_{cont} * VF); 1; lin$

RiRa → Filter

Item type: Connection

☐ Advanced filter treatment

Flow Rates

Filter Rates

Filter Release Rates

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←

ID	Start Time [h]	End Time [h]	Flow rate [1/h]
1 Default	t_end_leak	t_end	$= FF * Q_{vf} / (V_{cont} * VF); 1; lin$
2 iodine_organic	t_end_leak	t_end	$= FF_{or} * Q_{vf} / (V_{cont} * VF); 1; lin$
3 iodine_element	t_end_leak	t_end	$= FF_{el} * Q_{vf} / (V_{cont} * VF); 1; lin$
4 Default	t_start	t_end_leak	0.0
5 noble_gas	t_start	t_end	0.0

Figure 26: Filtering.